

Final Report on Olist Customer Satisfaction Analysis

NOTE: Except points 1, 2, 4, and 14, you can add/remove subpoints if necessary.

1. Name of the Project [1 mark]

Project Name	Olist Customer Satisfaction Analysis
Date	25-08-2026
Prepared By	Gayatri Behera
Executive Summary	This project analyses Olist customer satisfaction, identifies key drivers such as delivery delays and seller quality, and develops an XGBoost model to predict customer satisfaction and support targeted improvements.

2. Problem Statement [1 mark]

To identify the key factors affecting Olist customer satisfaction and develop a machine-learning model to predict satisfaction and support data-driven improvements in customer experience.

3. Project Goals and Objectives [1 mark]

Goal	Description	Criteria
Predict customer satisfaction	Build a machine learning model that predicts whether a customer will be satisfied or unsatisfied with their order, using order, delivery, seller, and product features.	Model achieves a meaningful improvement in ROC-AUC over the logistic regression baseline.
Identify key drivers of dissatisfaction	Use explainability techniques (SHAP and LIME) to determine which factors most influence predicted satisfaction, both globally and for individual orders.	Top drivers are identified and validated by at least two independent explanation methods.
Translate findings into business action	Map model drivers to plain-language business meaning and produce actionable recommendations for improving customer satisfaction.	Recommendations are directly traceable to specific model-identified drivers (e.g., delivery delay, seller quality).

4. Scope of the Project [1 mark]

In Scope	Analysis of Olist order, delivery, product, payment, and seller data to predict customer satisfaction. Includes data cleaning, exploratory data analysis, baseline and advanced model development (Logistic Regression, Random Forest, LightGBM, XGBoost), hyperparameter tuning, feature selection, and model explainability using SHAP and LIME. Also includes translating model drivers into business recommendations and building a summary dashboard of satisfaction patterns.
Out of Scope	Real-time or production deployment of the prediction model. Direct integration with Olist's live order-management or customer-service systems. Analysis of order data outside the provided dataset's time period. Root-cause investigation into individual seller operations or carrier-level logistics performance.

5. Assumptions [1 mark]

Assumption ID	Assumption
A1	The satisfied label, derived from review scores, is a reasonable proxy for actual customer satisfaction.
A2	The provided dataset is representative of Olist's broader order volume and is not significantly biased by how or when it was sampled.
A3	Delivery dates, estimated delivery dates, and review scores in the dataset are recorded accurately and consistently across orders.
A4	The relationships between delivery performance, seller quality, and satisfaction observed in this historical data will remain reasonably stable going forward.

6. Constraints [1 mark]

Constraint ID	Constraint	Description
C1	Historical, static dataset	The analysis is based on a fixed historical extract rather than a live data feed, so it cannot reflect very recent changes in delivery or seller behavior.
C2	Class imbalance	Unsatisfied orders make up a minority of the dataset (~21%), which limits how precisely the model can learn minority-class patterns and required imbalance-handling techniques.
C3	Limited hyperparameter search	Tuning was performed using a randomized search with a limited number of iterations (12) due to time and compute constraints, rather than an exhaustive grid search.

C4	Tool and environment limitations	Some explainability tooling (SHAP's TreeExplainer) had version-compatibility issues with the XGBoost version used, requiring a native-library workaround rather than the standard approach.
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7. Dependencies [1 mark]

Dependency ID	Dependency	Impact
D1	Availability and quality of the Olist dataset (orders, products, sellers, reviews, payments)	Any missing or corrupted source files would directly limit which features could be engineered and used in the model.
D2	Python libraries (pandas, scikit-learn, XGBoost, LightGBM, SHAP, LIME)	Model training, tuning, and explainability all depend on these libraries; version mismatches (as seen with SHAP/XGBoost) can block or delay analysis.
D3	Visualization tools (Tableau/Power BI)	The dashboard deliverable depends on these external tools to build and export charts for the report.
D4	Stage 2 baseline model and metrics	The comparative evaluation in this stage depends on the Logistic Regression baseline and metrics produced in the earlier project stage

8. Risks and Mitigation [1 mark]

Risk ID	Risk Description	Mitigation Strategy
R1	Model overfits to the training data (as seen with Random Forest's perfect training ROC-AUC)	Use stratified cross-validation and explicitly check the train-test/train-CV gap before selecting a final model; prefer models with smaller generalization gaps.
R2	Class imbalance causes the model to favor the majority (satisfied) class	Apply class-weighting (class_weight="balanced", scale_pos_weight) and evaluate using F1-macro and ROC-AUC rather than accuracy alone.
R3	Model predictions are not trusted or understood by business stakeholders	Use SHAP and LIME to produce global and local explanations, and translate top drivers into a plain-language business-mapping table.
R4	Library/version incompatibilities disrupt analysis (e.g., SHAP-XGBoost conflict)	Identify a workaround using native library functionality (XGBoost's built-in SHAP computation) and document the issue and fix for reproducibility.

9. Stakeholders (Stakeholder Analysis Matrix) [1 mark]

Stakeholder	Role	RACI	Interest and Influence
Product Manager (Analytics)	Owens the analysis, defines requirements, and translates findings into recommendations	A	HIGH
Data/ML Analyst (self)	Builds, tunes, and explains the predictive model	R	HIGH
Customer Experience / Support Team	Uses insights to prioritize interventions on at-risk orders	C	HIGH
Logistics/Operations Team	Responsible for delivery performance, a key driver identified by the model	C	MEDIUM
Seller Management Team	Responsible for seller quality, a key driver identified by the model	C	MEDIUM
Senior Leadership/Sponsor	Reviews outcomes and approves resourcing for recommended actions	I	LOW

10. Business Requirements (MoSCoW Prioritisation) [1 mark]

Priority	Requirement	Justification
MUST	Predict whether an order will result in a satisfied or unsatisfied customer	Core requirement — without this, the project cannot support proactive customer experience decisions.
MUST	Identify the key drivers of dissatisfaction using explainability techniques	Stakeholders need to understand <i>why</i> a prediction is made, not just the prediction itself, to act on it.

SHOULD	Quantify improvement over the existing baseline model	Demonstrates whether the added modeling effort (tuning, feature selection) delivers measurable value.
SHOULD	Provide business-friendly mapping of technical drivers to plain-language meaning and actions	Ensures non-technical stakeholders (support, logistics, seller teams) can act on the findings.
COULD	Build a summary dashboard of satisfaction patterns by state, category, and time	Supports exploratory monitoring and communication but is not essential to the core prediction task.
COULD	Export per-order predictions for downstream reporting	Useful for operational follow-up but not required to meet the core analytical objective.

11. Evaluation Metrics [1 mark]

Metric	Target	How It Is Measured	Achieved
ROC-AUC	Improve over Stage 2 baseline (0.6919)	Area under the ROC curve on the held-out test set	0.7062 (+0.0143 over baseline)
F1-macro	Outperform baseline given class imbalance	Macro-averaged F1 score across both classes on the test set	Tree models outperformed baseline (e.g., Random Forest F1-macro 0.6442 vs. baseline 0.6262)
Train–CV / Train–Test Gap	Keep generalization gap reasonably controlled	Difference between training and cross-validation/test ROC-AUC	Final model gap of 0.1120, flagged as large but improved via tuning/feature selection
Explainability Coverage	Produce both global and local explanations, validated by two methods	SHAP global summary, SHAP local waterfall, and LIME local explanation for the same order	Achieved — SHAP and LIME agreed on the top driver (seller reputation) for the example order

12. Sequence of Tasks Performed with Justification [2 marks]

(a) Data cleaning and preparation — Raw Olist order, product, seller, payment, and review data were merged, cleaned, and a satisfied target variable was derived from review scores. This step was necessary to produce a single reliable dataset (olist_clean.csv) that later stages could build on consistently.

(b) Exploratory data analysis and baseline model — Key patterns (e.g., delivery delay vs. satisfaction, satisfaction by state/category) were explored, and a Logistic Regression baseline model was trained. This established an interpretable reference point against which more advanced models could be measured.

(c) Advanced model comparison, tuning, and feature selection — Random Forest, LightGBM, and XGBoost were trained with class-imbalance handling and compared to the baseline using stratified cross-validation. XGBoost was selected, then tuned (RandomizedSearchCV) and reduced from 25 to 14 features, to address overfitting and improve generalization while quantifying the improvement over baseline.

(d) Explainability and reporting — SHAP (global and local) and LIME were applied to the final model to identify and validate key drivers of dissatisfaction, which were then translated into a business-mapping table and per-order prediction export, forming the basis for the final recommendations and dashboard.

13. Model Selection Summary [1 mark]

Model	Accuracy	Macro F1	ROC-AUC	Recall (unsatisfied)
Random Forest	0.8111	0.6442	0.7001	0.2997
LightGBM	0.7362	0.6353	0.7057	0.4985
XGBoost	0.7393	0.6372	0.7048	0.4958
Logistic Regression	0.7272	0.6262	0.6919	0.4921

Random Forest achieved the highest accuracy and Macro F1 but showed strong overfitting. LightGBM achieved the highest ROC-AUC (0.7057), while XGBoost achieved a very similar ROC-AUC (0.7048) with better cross-validation performance. XGBoost was therefore selected as the final tree model based on its highest 5-fold CV ROC-AUC of 0.7094 and a more reasonable bias–variance gap.

After hyperparameter tuning and feature selection, XGBoost improved test ROC-AUC from the Stage 2 baseline of 0.6919 to 0.7062, an improvement of 0.0143. The feature set was reduced from 25 to 14 features. Threshold optimisation further achieved 70.93% recall for the unsatisfied class at a threshold of 0.41, meeting the target recall of 70%.

14. Analytical Findings [1 mark]

(a) Key drivers of dissatisfaction – Delivery delay was the strongest overall driver of dissatisfaction, followed by seller average score and actual delivery time. Processing time, freight value, customer location frequency and product characteristics also influenced predictions. SHAP and LIME showed that seller quality can drive dissatisfaction even when an order is delivered early

(b) Limitations – The model is based on historical Olist data and may not generalise to future customer behaviour. Customer satisfaction is inferred from review scores rather than directly

measured, and the dataset is imbalanced between satisfied and unsatisfied orders. The model also has only modest predictive improvement over the baseline, so predictions should support rather than replace business decisions.

15. Business Recommendations [1 mark]

- (a) Reduce delivery delays:** Prioritise orders with high predicted dissatisfaction risk and focus on reducing delivery delays and total delivery time.
- (b) Monitor seller quality:** Track sellers with consistently low satisfaction scores and provide targeted support or corrective action.
- (c) Improve order processing:** Identify bottlenecks in processing and shipping so orders move faster even when they are still within the promised delivery date.
- (d) Target high-risk orders proactively:** Use the model's predicted probability to flag at-risk orders for early intervention, such as proactive customer communication or expedited resolution.

16. Glossary [1 mark]

Term	Definition
Customer Satisfaction	A binary outcome indicating whether a customer is satisfied or unsatisfied based on the review score
ROC-AUC	A metric that measures how well a model distinguishes between satisfied and unsatisfied customers across different classification thresholds.
Recall (Unsatisfied)	The proportion of actually unsatisfied orders that the model correctly identifies.
SHAP	An explainability method that shows how each feature contributes to an individual prediction and to the model overall.
Feature Selection	The process of retaining the most important features while removing less useful features to reduce model complexity and noise.

Project Deliverables:

- data_cleaning.ipynb
- eda_and_baseline_model.ipynb
- model_comparison_and_optimisation.ipynb
- dashboard.pdf (dashboard template doc converted to pdf)
- final_management_presentation.pptx (ppt saved as pdf)

- final_report.pdf (final report template doc converted to pdf)

Sign-off Status: Final analysis completed and ready for submission.